SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Duration : 1 Hour**

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I B.Deepthi -192424269 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B.Deepthi
Signature of the student:

1.WATER JUG PROBLEM (4-GALLON & 3-GALLON JUGS)

Problem Understanding

We have:

- Jug A = 4 gallons
- Jug B = 3 gallons

Allowed operations:

- Fill a jug completely.
- Empty a jug completely.
- Pour water from one jug to another until one jug is full or the other becomes empty.

Goal: Obtain exactly **2 gallons** in the **4-gallon jug**.

Method Selection

Use **State Space Search**.

Represent each state as

(4-gallon jug, 3-gallon jug)

Example:

(0,0)

means both jugs are empty.

Solution Steps

Step	Action	State (4G,3G)
1	Fill 3-gallon jug	(0,3)
2	Pour into 4-gallon jug	(3,0)
3	Fill 3-gallon jug again	(3,3)
4	Pour into 4-gallon jug until full	(4,2)
5	Empty 4-gallon jug	(0,2)
6	Pour remaining 2 gallons into 4-gallon jug	(2,0)

Final Answer

The **4-gallon jug contains exactly 2 gallons.**

Interpretation

The Water Jug problem is solved using **State Space Search**, where every possible jug configuration is treated as a state. Following legal operations leads to the goal state **(2,0)**.

2. MARS ROVER INTELLIGENT AGENT

i) Percepts

The rover receives information through its sensors.

Examples:

- Camera images
- Rock colour
- Soil composition
- Temperature
- Atmospheric pressure
- Radiation level
- Distance to obstacles
- Wheel position
- Battery level

ii) Operating Environment

Characteristic	Description
Observable	Partially Observable
Agents	Single Agent
Deterministic	Mostly Stochastic
Sequential	Yes
Dynamic	Yes
Continuous	Yes
Known	Partially Known

iii) Actions

The Mars Rover can

- Move Forward
- Move Backward
- Turn Left
- Turn Right
- Capture Images
- Drill Rocks
- Collect Soil Samples
- Analyze Samples
- Send Data to Earth
- Recharge using Solar Panels
- Avoid Obstacles

iv) Performance Measure

The rover performs well if it

- Collects maximum useful samples
- Avoids collisions
- Uses minimum energy
- Successfully sends data
- Completes mission quickly
- Travels safely

v) Best Agent Architecture

Utility-Based Learning Agent

Why?

Because the rover

- learns from experience,
- selects safest route,
- saves battery,
- adapts to unknown terrain,
- chooses the best action.

This makes it suitable for unpredictable Mars environments.

Interpretation

A Utility-Based Learning Agent provides intelligent decision-making while maximizing mission success and minimizing energy consumption.

3. 8-QUEENS PROBLEM

Problem Understanding

Place **8 queens** on an **8×8 chessboard** such that

- No two queens share a row
- No two queens share a column
- No two queens share a diagonal

Problem Components

Initial State

Empty chessboard

State Space

Every valid arrangement after placing queens.

Actions

Place one queen in the next column.

Goal Test

All 8 queens placed safely.

Path Cost

Every move costs **1**.

Search Strategy

Use **Backtracking Search**

Why?

- Easy
- Efficient
- Removes invalid positions immediately.

Simple Working

Place queen column-wise.

If conflict occurs

↓

Go back

↓

Try another row

↓

Repeat until solution.

One Valid Solution

Column	Row
1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

Interpretation

Backtracking avoids unnecessary exploration by rejecting invalid positions early, making it an efficient method for solving the 8-Queens problem.

4. OLA CAB BOOKING PROBLEM

Problem Understanding

A customer wants to travel.

The app provides

- Mini
- Micro
- Sedan

- Prime
- Shared

The system chooses the best cab according to user preference.

Problem-Solving Agent

Goal-Based Agent

Reach destination successfully.

The agent evaluates different options before selecting one.

Pseudocode

```
START

Input pickup location

Input destination

Display available cab types
Customer selects cab

IF cab available THEN

    Assign nearest driver

    Calculate fare

    Show ETA

    Confirm booking

ELSE

    Display "Cab Not Available"

ENDIF

Driver picks customer

Customer reaches destination

Payment completed
```

Interpretation

A Goal-Based Agent selects actions that satisfy the customer's objective of reaching the destination efficiently while considering availability and user preferences.

5. Uniform Cost Search (UCS)

(Your question mentions a weighted graph, but the graph image is missing. Below is the complete UCS method. You can replace the node costs with those from your graph.)

Problem Understanding

Goal:

Find the **minimum-cost path** from

S → G

using Uniform Cost Search.

Method Selection

Uniform Cost Search

Reason:

- Always expands the lowest cumulative cost node.
- Guarantees optimal solution.
- Suitable for weighted graphs.

UCS Algorithm

1. Insert Start node.
2. Remove lowest-cost node.
3. If Goal

found Return

solution. Otherwise

Expand neighbours.

Repeat.

Example

Suppose

S → A = 2

S → B = 4

A → C = 3

B → D = 2

C → G = 4

D → G = 1

UCS Table

Expanded	Cost
S	0
A	2
B	4
C	5
D	6
G	7

Least Cost Path

S

↓

A

↓

C

↓

G

Total Cost

2 + 3 + 2 = 7

Interpretation

Uniform Cost Search always expands the least-cost node first, ensuring that the final path is the optimal route with the minimum transportation cost.

Conclusion

Question	Method Used	Reason
1. Water Jug	State Space Search	Finds goal state systematically
2. Mars Rover	Utility-Based Learning Agent	Learns and adapts in dynamic environments
3. 8 Queens	Backtracking Search	Efficiently avoids invalid placements
4. OLA Cab	Goal-Based Agent	Selects actions to reach the destination
5. Logistics Network	Uniform Cost Search (UCS)	Finds the least-cost path in weighted graphs